

BK8520 Electronic Load REST API Flow Diagram

This Mermaid diagram illustrates the POST and GET API endpoints for the BK8520 Electronic Load Control system.


```

graph TB
    subgraph "Client Applications"
        WEB[Web Interface]
        PYTHON[Python Client]
        CURL[cURL/CLI]
        JS[JavaScript Client]
    end

    subgraph "BK8520 REST API Server"
        subgraph "Health & System"
            HEALTH["GET /api/health"]
            Check service status
            Returns: API health info"
        end

        subgraph "Device Management"
            CONNECT["POST /api/device/connect"]
            Connect to BK8520
            Params: port, baudrate
            Action: Establish serial connection"
            DISCONNECT["POST /api/device/disconnect"]
            Disconnect device
            Params: none
            Action: Close serial connection"
            STATUS["GET /api/device/status"]
            Get connection status
            Returns: connection state, device status"
            INFO["GET /api/device/info"]
            i Get device details
            Returns: model, firmware, capabilities"
            READINGS["GET /api/device/readings"]
            ✂ Get live measurements
            Returns: voltage, current, power"
        end

        subgraph "Device Control"
            INPUT["POST /api/device/input"]
            ✂ Control load input
            Params: enabled (true/false)
            Action: Turn input on/off"
            CURRENT["POST /api/device/current"]
            ✂ Set discharge current
            Params: current (amperes)
            Action: Configure CC mode current"
            VOLTAGE["POST /api/device/voltage"]
            ✂ Set cutoff voltage
            Params: voltage (volts)
            Action: Set battery protection limit"
            MODE["POST /api/device/mode"]
            ⚙ Set operation mode
            Params: mode (0=CC,1=CV,2=CW,3=CR)
            Action: Change load operation mode"
            MAXCURRENT["POST /api/device/max-current"]
            Set current limit
        end
    end

```

```

Params: current (amperes)
Action: Configure safety limit"]
        MAXPOWER["POST /api/device/max-power
    Set power limit
Params: power (watts)
Action: Configure safety limit"]
        MAXVOLTAGE["POST /api/device/max-voltage
    Set voltage limit
Params: voltage (volts)
Action: Configure safety limit"]
        SETUP["POST /api/device/setup-discharge
❖ Setup CC discharge
Params: current (amperes)
Action: Enable constant current mode"]
        end

        subgraph "Battery Testing"
            TESTSTART["POST /api/battery-test/start
        Start capacity test
Params: discharge_current, cutoff_voltage
Action: Begin automated test"]
            TESTSTOP["POST /api/battery-test/stop
        Stop active test
Params: none
Action: Halt test, save results"]
            TESTSTATUS["GET /api/battery-test/status
        Get test progress
Returns: test state, measurements
Updates: capacity, elapsed time"]
            TESTRESULTS["GET /api/battery-test/results
        Get complete results
Returns: full test data
Includes: capacity, duration, summary"]
        end
    end

    subgraph "BK8520 Hardware Device"
        DEVICE["BK8520 Electronic Load
    Serial Communication
Port: /dev/ttyUSB0
Baudrate: 4800
Max: 120V, 60A, 999W"]
    end

    %% Client connections
    WEB --> HEALTH
    WEB --> CONNECT
    WEB --> STATUS
    WEB --> READINGS
    WEB --> INPUT
    WEB --> TESTSTART
    WEB --> TESTSTATUS

    PYTHON --> CONNECT
    PYTHON --> CURRENT

```

```
PYTHON --> VOLTAGE
PYTHON --> TESTSTART
PYTHON --> TESTRESULTS
```

```
CURL --> HEALTH
CURL --> INFO
CURL --> MODE
```

```
JS --> READINGS
JS --> TESTSTATUS
```

```
%% API to Device connections
```

```
CONNECT --> DEVICE
STATUS --> DEVICE
INFO --> DEVICE
READINGS --> DEVICE
INPUT --> DEVICE
CURRENT --> DEVICE
VOLTAGE --> DEVICE
MODE --> DEVICE
MAXCURRENT --> DEVICE
MAXPOWER --> DEVICE
MAXVOLTAGE --> DEVICE
SETUP --> DEVICE
TESTSTART --> DEVICE
TESTSTOP --> DEVICE
TESTSTATUS --> DEVICE
```

```
%% Styling
```

```
classDef getEndpoint fill:#e1f5fe,stroke:#0277bd,stroke-width:2px
classDef postEndpoint fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px
classDef device fill:#fff3e0,stroke:#f57c00,stroke-width:3px
classDef client fill:#e8f5e8,stroke:#388e3c,stroke-width:2px
```

```
class HEALTH,STATUS,INFO,READINGS,TESTSTATUS,TESTRESULTS getEndpoint
class CONNECT,DISCONNECT,INPUT,CURRENT,VOLTAGE,MODE,MAXCURRENT,MAXPOWER,MAXVOLTAGE,SE
class DEVICE device
class WEB,PYTHON,CURL,JS client
```

Note: This diagram would be rendered visually in a Mermaid-compatible viewer

API Endpoint Details

GET Endpoints (Data Retrieval)

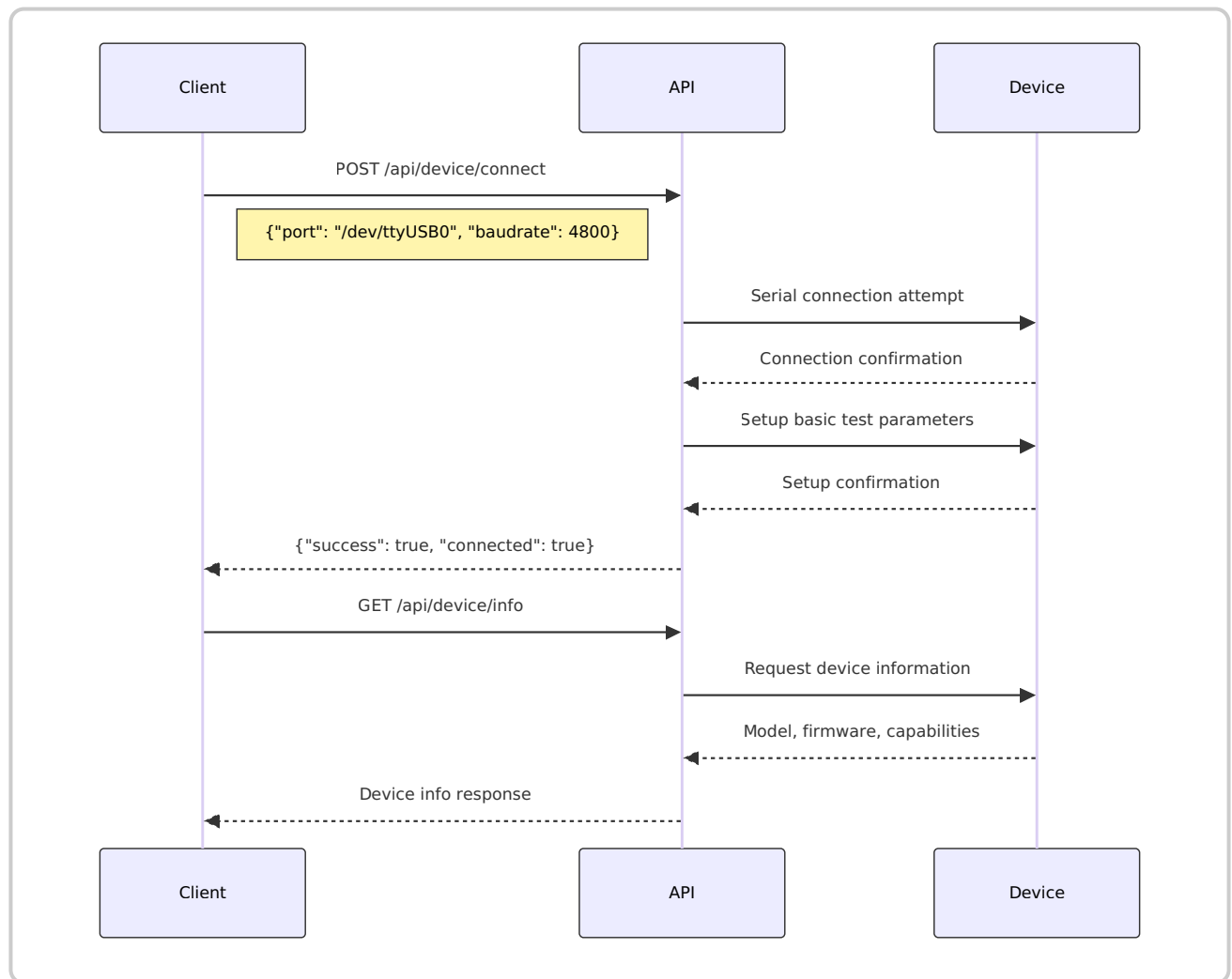
Endpoint	Purpose	Response Data
GET /api/health	Service health check	Status and version info
GET /api/device/status	Connection and operational status	Connection state, device status
GET /api/device/info	Device information	Model, firmware, capabilities
GET /api/device/readings	Real-time measurements	Voltage, current, power values
GET /api/battery-test/status	Active test status	Test progress, measurements
GET /api/battery-test/results	Complete test results	Full test data and summary

POST Endpoints (Control Operations)

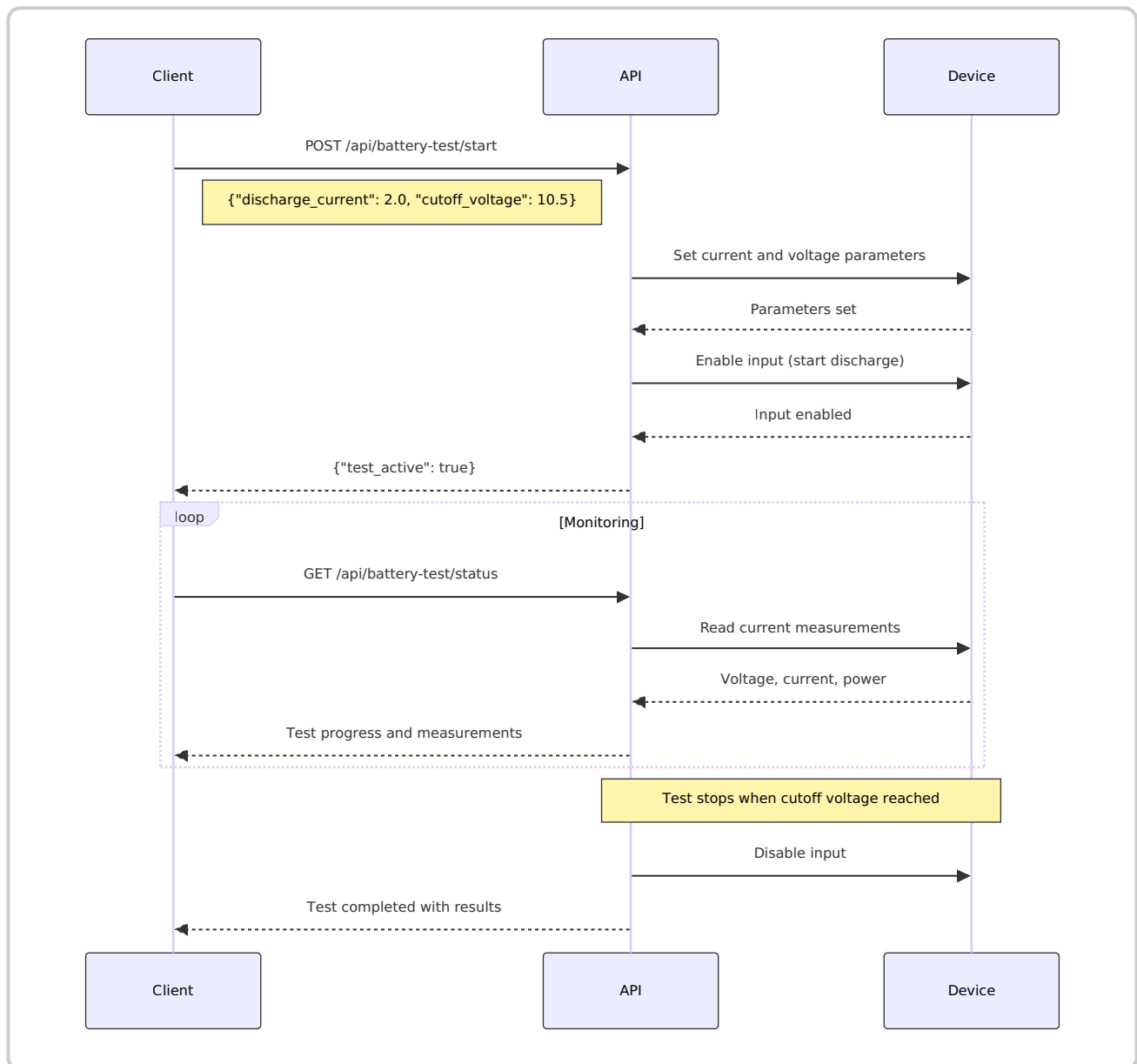
Endpoint	Purpose	Parameters
POST /api/device/connect	Connect to device	Port, baudrate, timeout
POST /api/device/disconnect	Disconnect device	None
POST /api/device/input	Control input on/off	enabled (boolean)
POST /api/device/current	Set discharge current	current (amperes)
POST /api/device/voltage	Set cutoff voltage	voltage (volts)
POST /api/device/mode	Set operation mode	mode (0=CC, 1=CV, 2=CW, 3=CR)
POST /api/device/max-current	Set current limit	current (amperes)
POST /api/device/max-power	Set power limit	power (watts)
POST /api/device/max-voltage	Set voltage limit	voltage (volts)
POST /api/device/setup-discharge	Setup CC discharge	current (amperes)
POST /api/battery-test/start	Start battery test	discharge_current, cutoff_voltage, max_time_hours
POST /api/battery-test/stop	Stop battery test	None

Data Flow Examples

Device Connection Flow



Battery Test Flow



System Architecture

The API server acts as a bridge between client applications and the physical BK8520 device:

- **FastAPI Server:** Handles HTTP requests and provides OpenAPI documentation
- **Serial Communication:** Direct communication with BK8520 via RS232/USB
- **Real-time Monitoring:** Continuous measurement reading during tests
- **Data Persistence:** Test results saved to JSON files
- **Error Handling:** Comprehensive error reporting and device safety checks

Base URL: <http://10.100.10.190:8000>